Lab-Network Address Translation (NAT)

Objectives:

- To distinguish between Inter-networking using routers versus Network Address Translation (NAT)).
- To distinguish between private and public IP addresses
- To acquire hands-on working knowledge of Internet Packets, Protocol, and Addresses.

Exercises

Exercise 1 – Hosts & Network Devices

Physical and Virtual DHCP & NAT function Physical DHCP & NAT

- These exercises can be done on the Ubuntu partition, or on the Windows partition with a Linux virtual machine.
- First, we need to appreciate the difference between a private IP address and a public IP address. ICANNA/IANA has allocated certain range of IP addresses for private local use (see Appendix A). These addresses are not allowed for use on the public networks. That is, the routers in the public networks are NOT configured to route them.
- Being private limits their use to only private networks. They are re-used extensively. For example, each Wireless Access Point (AP) serving a home or a small business, typically use the default IP address range 192.168.1.0/24 (256 IP addresses). How many laptops and desktops around the world uses or assigned 192.168.1.64/24? Millions! If they are allowed to be routable, then it will be a total chaos.
- Before, we proceed; let us get acquainted with a tool that would enable you to determine if an IP address is private or public. That is, **ipcalc** on Linux.
- Open a Linux terminal on your Ubuntu Host or Linux Mint VM and type

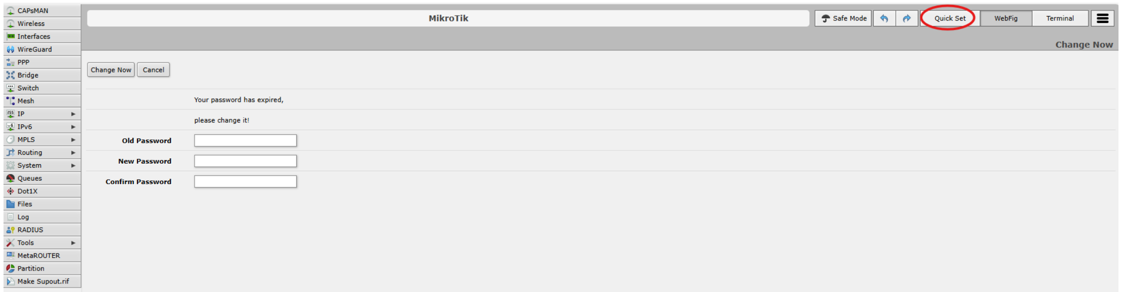
```
ipcalc 192.168.1.12/24.
```

- If you receive an error that ipcalc is not installed, you should run “sudo apt install ipcalc”, enter the host password, and then agree to any conditions.
- Note that /24 is a short hand for 255.255.255.0 (it represents a subnet mask of 24 1s). Also, note that 255 in binary is 1111 1111. The response should look like Figure 1 below. As you can see in Figure 1, 192.168.1.122/24 is a private IP address. If the “Private Internet” is not there, then the IP address in public IP address

```
emil@precision-m4700:~/.mozilla/firefox$ ipcalc 192.168.1.122/24
Address:   192.168.1.122        11000000.10101000.00000001. 01111010
Netmask:   255.255.255.0 = 24   11111111.11111111.11111111. 00000000
Wildcard:  0.0.0.255            00000000.00000000.00000000. 11111111
=>
Network:   192.168.1.0/24       11000000.10101000.00000001. 00000000
HostMin:   192.168.1.1          11000000.10101000.00000001. 00000001
HostMax:   192.168.1.254        11000000.10101000.00000001. 11111110
Broadcast: 192.168.1.255        11000000.10101000.00000001. 11111111
Hosts/Net: 254                   Class C, Private Internet
```

Figure 1 – ipcalc response for a private ip address

  - Q1 – Determine whether 134.126.10.64/22 is private or public IP address?
- Let us learn about two major functionalities/devices used extensively in Home and small business LAN networks. These are NAT (Network Address Translation) and private DHCP (Dynamic Host Configuration Protocol) server configured for private IP address dynamic assignment.
  - You will need to pick up an Mikrotik Access Point (AP), known also in layman terms router or wireless router, from the lab manager. Power on the AP and your laptop.
  - An AP has a unique functionality called NAT (Network Address Translation). It allows the AP to connect two networks only.
  - Typically a private network (with multiple hosts) to A SINGLE public IP address. This functionality is known as NAT. NAT allows a single device, such as AP, to act as an agent between the Internet (or public network) and a local network (or private network), which means that only a single unique IP address is required to represent an entire group of computers to anything outside their network. Because APs inter-network two networks, the general public erroneously refer to them as routers. But they are not routers; they are Access Points.
- Using an Ethernet cord/cable, connect the Ethernet port on the laptop to one of the LAN ports (2/3/4) on the front of the AP. You will need to disconnect and connect the Ethernet connection on Ubuntu Desktop to acquire an IP address from the AP.
  - Check that Ubuntu acquired an IP address using the command "ip address". Make sure that the IP address has the following first three segments 192.168.88.X followed with an integer from 2 to 254 (for example, 192.168.88.183).
  - **Q2** – State the IP address acquired by your machine and identify the server in the AP that allows the Ethernet port to acquire an IP address.
- Log into the AP by launching a browser on your laptop and type the following in the URL field: 192.168.88.1. You should be presented with the AP Home page. Click on the Setup/QuickSet tab and you will be presented with a window for authentication. If you are not automatically logged in, the username is admin and there is no password.

- 

- In the Local Network subsection (middle right), change the IP Address into 10.<section number>.<last 2 digits of ID number>.1. For example, if you are in section 1 and group 5, then it is 10.2.5.1.
- 

    - In the same subsection, change the DHCP Server Range to one that matches the IP address you used previously. Continuing with the example of 10.2.5.1, you should set the DHCP range to 10.2.5.10-10.2.5.20.
    - Scroll to the bottom of the page and click Apply Configuration. After, refresh the page. Leave this for around a minute, then power the device off by pulling the power cable out, leave this powered off for 10 seconds, then power the device back on.
    - After restarting, you will need to navigate to the new IP address to access this page again.
    - **Q3** – Is 10.2.5.1 public or private IP address? Provide evidence in support of your claim.
- Find the Wireless subsection (Top Left). Change the Network Name under 2GHz to Sec<your section number>Grp<your group number>. Continuing with the example above, the SSID should be Sec2Grp5. Click Save at the bottom of the page.
    - Change the Frequency under 2GHz to the channel number corresponding to your group number. Continuing the example, one would select 5 – 2.432 GHz.
    - Click “Apply Configuration”. You can then look for the wireless access point on the WiFi list of the host machine. If you cannot see it, reset the Mikrotik AP in the same way we did prior.
    - **Q4** – What is the new IP acquired by Ubuntu?

- The next step is to check whether we have connectivity to laptops in other groups. Open a terminal on the host Ubuntu Desktop and ping the IP address acquired by another group in the class. That is, type

ping –c 5 <ip address of the other group>

Hint: This should fail because we do not have connection to their network.

- Let us fix this by unplugging from your AP and connecting to their AP (Wired or wireless connection). Let us find the IP address acquired by the Ethernet Port.
- Now ping the laptop of the other group from yours using their IP address, type

  ping –c 5 <ip address of the other group>

  - **Q5** – Is the ping (ICMP request) successful? Explain.

Ping one of University of Bristol's server at 137.222.8.142 by typing

ping –c 6 137.222.8.142

in a terminal on your laptop.

  - **Q6** – Is the ping successful? Explain.

- Disconnect from the other group's AP and connect to yours. Connect another Ethernet cable from the WAN (First port on the left) on the AP to the main AP in the lab. Now try to ping the server again.
  - **Q7** – Is the ping (ICMP request) successful? Explain.

Virtual DHCP and NAT

- The functionality of Private IP DHCP server and NAT that are available on an AP is also available on a virtual basis in VMware Workstation and it is known as vmnet8 virtual switch.
- So, when you configure the network adapter on, say, an Ubuntu VM to NAT connection type, you are connecting the VM to the virtual device vmnet8. It is the same as connecting your laptop to one the physical AP yellow port. Vmnet8 has a Private IP DHCP server and NAT functionality built-in. VMware vmnet8 virtual switch WAN port is automatically connected to the Ethernet port of the Host. That is equivalent to connecting the Blue WAN port of the AP to the switch.
- Set up Ubuntu VM to be connected to vmnet8 by configuring its Network Adapter for NAT connection.
  - **Q8** - Check out its connectivity to the Host IP address and the JMU email server. Hint: ping.
  - **Q9** – Draw and submit a network diagram including Ubuntu VM, vmnet8 (virtual Private IP DHCP server, virtual NAT), and Host.

Additional Questions (beyond those under the exercises)

Q1 What is the difference between Dynamic and Static IP addresses?

Appendix A: Private IP Addresses

- Private Addresses
    - 10.0.0.0 – 10.255.255.255 (16,777,216) – Single Class A (24 bits)
    - 172.16.0.0 – 172.31.255.255 (1,048,576) – 16 contiguous class Bs (20 bits)
    - 192.168.0.0 – 192.168.255.255 (65,536) – 256 contiguous class Cs (16 bits)
- Link-local addresses

If a host on an IEEE 802 (Ethernet) network cannot obtain a network address via DHCP, an address from 169.254.1.0 to 169.254.254.255 may be assigned pseudo-randomly. Windows OS uses 169.254.1.0 to 169.254.254.255 range. However, Linux OS may use 10.0.0.0 range.

Cheat sheet – LINUX CLI

Please add as you deem appropriate…

Command	**Brief Description**
cd	Change directory
ls	List file and folder in the current directory
ls -la	Same as ls with details to do with ownership and permissions time size, etc.
sudo vmware	To launch VMware workstation
sudo	Super user do
pwd	Current directory and its path relative to root
chown	Change ownership
chmod	Change permission
nano	Text editor
gedit	Graphical text editor
mkdir	Create a directory
rmdir	Remove directory
cp	Copy file
service	Service <service> restart
more	Display the content of a text file
cat	Display the content of a text file or multiple files
apt-get	apt-get install <package>
ifconfig	Display the details of network interfaces
ssh	Establish an ssh secure connection
man	Provide the manual (help) page of a command or utility

For detailed description (a manual) type **man** <the command>

Host Ubuntu 24.04: checkout, hellocheckin

Guest Windows 11: checkout, checkout

Guest Ubuntu 24.04: checkout, checkout

iMac: admin, 3022$3022